

Inputs

1. EgoClip narrations (pretraining corpus)

cook rice
cut onion
wash plate
...

2. Exposure parsing and aggregation

verb freq. $f_v(v)$
noun freq. $f_n(n)$
pair freq. $f_p(v, n)$
PMI(v, n)

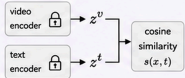
3. EK100 query example

video x



caption t : "cut onion"

4. Frozen EgoVLPv2 (dual-encoder)



Diagnostic branches (frozen representations)

(A) Exposure branch

Lookup exposure for the query pair (v, n)

Categorical split

SC

(seen pair)

UC

(unseen composition)

UA

(unseen atom)

Continuous exposure

$$\ell_p = \log_{10}(1 + f_p(v, n))$$

(B) Binding probe branch

Construct controlled captions

Positive (P)

cut onion

Noun swap (N^-)

cut plate

Verb swap (V^-)

wash onion

Compute
similarities
 $s(x, \cdot)$

s^+

s^{N^-}

s^{V^-}

noun margin
 $m^n = s^+ - s^{N^-}$

verb margin
 $m^v = s^+ - s^{V^-}$

Outputs (diagnostic measurements)

1. Retrieval performance

mAP, nDCG
($V \rightarrow T$ and $T \rightarrow V$)

2. Binding behaviour

Noun-swap accuracy
Verb-swap accuracy
Margins: m^n , m^v

3. Exposure dependence

Correlations (Pearson r / Spearman ρ
and exposure-controlled regression)

- $\rho(\text{retrieval}, \ell_p)$
- $\rho(m^n, \ell_p)$
- $\rho(m^v, \ell_p)$